

01_backtest_run

May 9, 2026

1 01 — Rolling-Window Backtest on Raw Prices

This notebook is the starting point of the benchmark. The numbered list below is the table of contents; it roughly tracks how we worked through the problem.

1. **Environment setup.** Imports, seeds, output directories.
2. **Obtain the data.** ICE *Coffee C* .xls from ice.com/report/293 (Feb 2026). For simplicity we work with a single series — the **nearby close** (front-month, most actively traded) — saved as `data/coffee.csv`.
3. **Assemble the 10-model suite.** Five categories — simple baselines, classical statistical, volatility, structural, and foundation models — selected to span a range of approaches and complexity.
4. **First test: a single forecast.** Hold out the most recent 63 trading days; each model forecasts them from the prior 1,536-day context. IBM Granite TTM topped this run, which fit our initial expectation that the more sophisticated models would dominate. The multi-origin results in §6 turn out differently.
5. **Forecast origin coverage.** One window isn't enough — we needed multiple origins distributed across the 32-year history. The 1, 10, 30, 60 sequence is the iteration trail: 10 too sparse, 30 better, 60 settled.
6. **Run the rolling-window backtest.** Scale by scale (10, then 30, then 60), checking results at each step rather than firing everything at once.
7. **Export the two CSVs** every downstream notebook consumes.
8. **The 60-origin leaderboard.** The final ranking from the scale we settled on — and the surprise (RWD on top, Granite mid-pack) that drives everything after.
9. **MAE across forecast-origin scales.** How mean MAE changes as we add origins, supporting our choice of 60.
10. **MAE distribution at 60 origins.** Boxplots sorted by median with the mean overlaid; two-tone coloring separates the statistically-tied cluster from Prophet, the lone MCS outlier.
11. **Per-origin deep dive: IBM Granite TTM vs. RWD.** How the foundation model loses: occasional collapse or steady underperformance?

File	Rows	Purpose
<code>results/csv/summary_all_scale10.csv</code>	1,010	One row per (origin, model); MAE / RMSE / MAPE / sMAPE / MASE
<code>results/csv/step_errors_all_scale10sx</code>	63 = 63,630	One row per (scale, model, origin, step); absolute error at each forecast step

The step-errors table is the input to the Diebold-Mariano test and the Model Confidence Set in notebook 02.

1.1 Backtest design

- **Context window:** 1,536 trading days (≈ 6 years). Granite TTM offers several zero-shot context-length variants (512, 1024, 1536); we picked the largest to give every model as much history as possible while keeping the comparison fair across the suite.
- **Horizon:** 63 trading days (≈ 3 months), a middle ground between a horizon too short to be interesting and one too long to be defensible.
- **Scales:** 1, 10, 30, 60 evenly-spaced forecast origins. 60 is the one we settled on; the rest are the trial-and-error that got us there.
- No model sees test data during fitting; every origin gets a fresh fit on the preceding 1,536 days.

1.2 1. Environment setup

```
[1]: import logging
import random
import sys
from pathlib import Path

# Make `coffee_forecast` importable whether or not `pip install -e .`
# is in effect for this kernel. No-op when the package is already on
# sys.path (e.g. via the editable install); falls back to inserting
# the repo root so the notebook also runs from a fresh clone.
try:
    import coffee_forecast # noqa: F401
except ModuleNotFoundError:
    sys.path.insert(0, str(Path.cwd().parent))

# Mute the noisiest third-party libraries.
for name in ("cmdstanpy", "prophet", "statsforecast"):
    logging.getLogger(name).setLevel(logging.ERROR)

import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import torch

from coffee_forecast import (
    load_coffee_data,
    run_multi_scale_backtest,
    StatsForecastWrapper,
    GARCHWrapper,
    ProphetWrapper,
    MLRecursiveWrapper,
    GraniteWrapper,
```

```

)
from coffee_forecast.backtest import step_errors_to_long_df
from coffee_forecast.config import (
    CONTEXT_LEN,
    HORIZON,
    SCALES,
    SEED,
    CSV_DIR,
    FIG_DIR,
    REPO_ROOT,
)
from coffee_forecast.viz import plot_price_series, plot_forecast_origin_coverage

# Seed every stochastic component we touch directly. Most of the suite
# (statsforecast MLE, arch MLE, Granite TTM, Prophet MAP) is deterministic
# given fixed inputs; RandomForest takes ``random_state=SEED`` at
# construction (see §3). Setting these global seeds guards against any
# incidental randomness in feature construction or library internals.
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)

# Ensure output directories exist.
CSV_DIR.mkdir(parents=True, exist_ok=True)
FIG_DIR.mkdir(parents=True, exist_ok=True)

```

1.3 2. Source the data

The raw `.xls` from ice.com/report/293 is a single sheet with 43 columns — the nearby contract plus four later-dated expiries, each carrying `CONTRACT` (YYYY/MM), `HIGH`, `LOW`, `CLOSE`, `OPEN`, `INTEREST`, `VOLUME`, with a couple of volatility series tacked on at the end — and four header rows above the data. The only column we use is the **nearby close**, so the one-time script below strips the file down to two columns and saves the result as `data/coffee.csv`. After that, every notebook and script imports the `.csv` directly.

The script is `scripts/export_csv.py`:

```

df = pd.read_excel("data/Coffee_Historical_Prices.xls", engine="xlrd")
df = df.iloc[4:, [0, 9]] # skip header rows; keep DATE + nearby CLOSE
df = df.rename(columns={df.columns[0]: "ds", df.columns[1]: "y"})
df["ds"] = pd.to_datetime(df["ds"])
df["y"] = pd.to_numeric(df["y"])
df = df.dropna(subset=["y"]).reset_index(drop=True)
df.to_csv("data/coffee.csv", index=False)

```

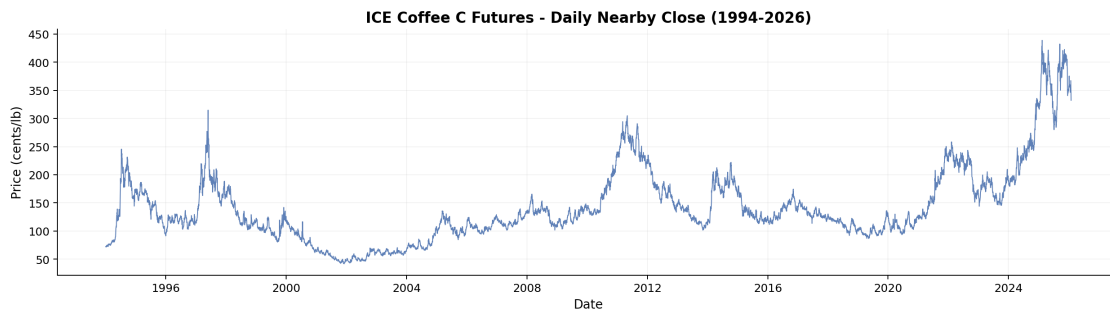
`load_coffee_data` in `coffee_forecast/data.py` is the thin reader the notebook calls: it loads the `.csv`, parses the date column, normalizes timestamps to midnight, sorts by date, and adds the `unique_id` column `statsforecast` requires.

```
[2]: df = load_coffee_data()
print(
    f"Loaded {len(df)} trading days ({df['ds'].min().date()} → {df['ds'].max().
    ↪date()})"
)
df.head()
```

Loaded 8104 trading days (1994-01-03 → 2026-01-30)

```
[2]:      ds      y unique_id
0 1994-01-03  72.50    coffee
1 1994-01-04  71.85    coffee
2 1994-01-05  71.55    coffee
3 1994-01-06  72.10    coffee
4 1994-01-07  72.50    coffee
```

```
[3]: _ = plot_price_series(df, save_path=FIG_DIR / "price_series.png")
plt.show()
```



1.4 3. Assemble the 10-model suite

Every wrapper exposes a uniform `predict(context_df, horizon) -> ndarray` interface, so the backtest runner treats them all identically.

Category	Models
Simple baselines	Naive, RWD (random walk with drift), Theta
Classical statistical	ETS (AutoETS), ARIMA (AutoARIMA)
Volatility	GARCH(1,1) with AR(1) mean
Structural decomposition	Prophet
Recursive ML	Ridge, Random Forest (lag + rolling features)
Foundation (zero-shot)	IBM Granite TTM

GARCH note. This is the simple GARCH(1,1)-Normal spec on raw prices, used here for its point forecast. Notebook 05 deploys a richer GJR-GARCH(1,1)-t spec on log-returns; that variant lives there because deployment is about prediction intervals, where the volatility specification does the work, not point forecasts.

```

[4]: from sklearn.ensemble import RandomForestRegressor
from sklearn.linear_model import Ridge
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from statsforecast.models import (
    AutoARIMA,
    AutoETS,
    Naive,
    RandomWalkWithDrift,
    Theta,
)
from tsfm_public.toolkit.get_model import get_model

device = "cuda" if torch.cuda.is_available() else "cpu"

ttm = (
    get_model(
        model_path="ibm-granite/granite-timeseries-ttm-r2",
        model_name="ttm",
        context_length=CONTEXT_LEN,
        prediction_length=96, # fixed by this Granite TTM variant; trimmed to
        ↪HORIZON in GraniteWrapper
        prefer_longer_context=True,
        return_model_key=False,
    )
    .to(device)
    .eval()
)
print(f"IBM Granite TTM loaded on {device.upper()}")

models = {
    # Simple baselines
    "Naive": StatsForecastWrapper(Naive(alias="Naive")),
    "RWD": StatsForecastWrapper(RandomWalkWithDrift(alias="RWD")),
    "Theta": StatsForecastWrapper(Theta(season_length=252, alias="Theta")),
    # Classical
    "ETS": StatsForecastWrapper(AutoETS(alias="ETS")),
    "ARIMA": StatsForecastWrapper(AutoARIMA(seasonal=False, alias="ARIMA")),
    # Volatility
    "GARCH": GARCHWrapper(),
    # Structural decomposition
    "Prophet": ProphetWrapper(),
    # Recursive ML
    "Ridge": MLRecursiveWrapper(make_pipeline(StandardScaler(), Ridge())),
    "RF": MLRecursiveWrapper(
        RandomForestRegressor(n_estimators=100, random_state=SEED)
    ),
}

```

```

# Foundation model
"Granite-TTM": GraniteWrapper(ttm, device, context_len=CONTEXT_LEN),
}
print(f"{len(models)} models initialized across 5 categories.")

```

```

INFO:p-89106:t-8478283968:get_model.py:get_model:Loading model from: ibm-
granite/granite-timeseries-ttm-r2
INFO:p-89106:t-8478283968:get_model.py:get_model:Model path type is 2
INFO:p-89106:t-8478283968:get_model.py:get_model:Attempting model load from ibm-
granite/granite-timeseries-ttm-r2, revision = 1536-96-r2.
INFO:p-89106:t-8478283968:get_model.py:get_model:Model loaded successfully from
ibm-granite/granite-timeseries-ttm-r2, revision = 1536-96-r2.
INFO:p-89106:t-8478283968:get_model.py:get_model:[TTM] context_length = 1536,
prediction_length = 96

IBM Granite TTM loaded on CPU
10 models initialized across 5 categories.

```

1.5 4. First test: a single forecast

We started by forecasting straight into the future. The outputs looked reasonable, but without realized prices to compare against we couldn't compute any error metrics. So we pulled the test window back to the most recent 63 trading days, where real prices already existed. Each model saw the 1,536 days before that as context and predicted the 63 days we already knew. A natural first choice of window, but enough to see real forecasts plotted against real prices for the first time. The metrics from this single window are saved as `single_origin_metrics.csv`.

```

[5]: from coffee_forecast import calculate_metrics, get_forecast_origins
    from coffee_forecast.config import COLOR_MAP, PLOT_ORDER

# Same machinery as the rolling-window backtest at scale=1: the latest
# valid forecast origin in the series.
a = get_forecast_origins(df, num_windows=1)[0]
context_df = df.iloc[a - CONTEXT_LEN : a].copy()
test_df = df.iloc[a : a + HORIZON].copy()
y_true = test_df["y"].to_numpy(dtype=float)
train_y = context_df["y"].to_numpy(dtype=float)
test_start = test_df["ds"].iloc[0]

# Generate forecasts.
plot_df = pd.DataFrame({"ds": test_df["ds"].to_numpy(), "Actual": y_true})
for name, wrapper in models.items():
    plot_df[name] = wrapper.predict(context_df, HORIZON)
    print(f"    {name}")

# Single-origin metrics, sorted by MAE.
rows = []
for name in PLOT_ORDER:

```

```

y_hat = plot_df[name].to_numpy(dtype=float)
metrics = calculate_metrics(y_true, y_hat, train_y)
metrics["Model"] = name
rows.append(metrics)

holdout_metrics = (
    pd.DataFrame(rows)
    .sort_values("MAE")
    .reset_index(drop=True)[["Model", "MAE", "RMSE", "sMAPE", "MASE"]]
)
holdout_metrics.to_csv(CSV_DIR / "single_origin_metrics.csv", index=False)
print("\nSaved: results/csv/single_origin_metrics.csv")
holdout_metrics.round(3)

```

Naive
RWD
Theta
ETS

ERROR:p-89106:t-8478283968:plot.py:<module>:Importing plotly failed. Interactive plots will not work.

ARIMA
GARCH
Prophet
Ridge
RF
Granite-TTM

Saved: results/csv/single_origin_metrics.csv

```

[5]:
  0  Granite-TTM  20.875  22.601  0.055  5.781
  1      GARCH  23.348  25.852  0.062  6.466
  2      RF    24.552  26.087  0.065  6.800
  3    Ridge  24.709  29.209  0.065  6.843
  4      ETS  26.039  29.876  0.069  7.212
  5    Naive  26.039  29.876  0.069  7.212
  6    ARIMA  26.044  29.927  0.069  7.213
  7    Theta  27.325  31.971  0.072  7.568
  8      RWD  29.231  35.043  0.077  8.096
  9    Prophet  49.243  60.306  0.124 13.638

```

Granite TTM tops the leaderboard on this single window, which made us optimistic moving forward. The multi-origin backtest in §6 turns the story upside down: RWD, second-to-last on this single window, ends up the mean-MAE leader across 60 origins.

```

[6]: # Show the last 63 trading days *before* the holdout for context
VIS_DAYS = HORIZON

```

```

vis_start = max(0, a - VIS_DAYS)
vis_df = df.iloc[vis_start : a + HORIZON][["ds", "y"]].copy()

fig, ax = plt.subplots(figsize=(10, 4), dpi=180)

ax.axvspan(test_start, vis_df["ds"].max(), color="gray", alpha=0.08)
ax.plot(
    vis_df["ds"], vis_df["y"], color=COLOR_MAP["Actual"], linewidth=1.4,
    label="Actual"
)

for name in PLOT_ORDER:
    if name not in plot_df.columns:
        continue
    y_hat = plot_df[name].to_numpy(dtype=float)
    if np.any(~np.isfinite(y_hat)):
        continue
    ax.plot(
        plot_df["ds"],
        y_hat,
        color=COLOR_MAP.get(name, "#666666"),
        linewidth=1.5,
        alpha=0.88,
        label=name,
    )

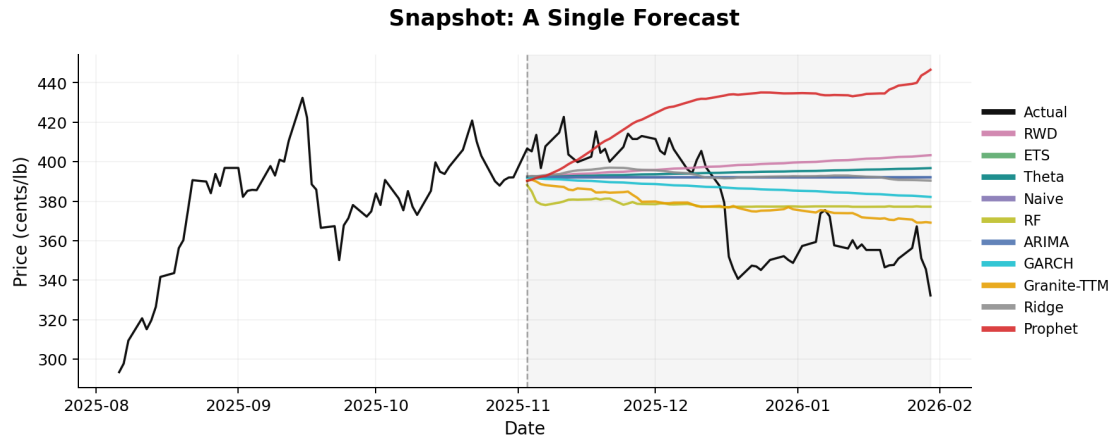
ax.axvline(test_start, ls="--", linewidth=1, color="black", alpha=0.35)

fig.suptitle("Snapshot: A Single Forecast", fontsize=14, weight="semibold", y=0.
    98)
ax.set_xlabel("Date", fontsize=11)
ax.set_ylabel("Price (cents/lb)", fontsize=11)
ax.grid(True, alpha=0.15)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

leg = ax.legend(loc="center left", bbox_to_anchor=(1, 0.5), frameon=False,
    fontsize=9)
for line in leg.get_lines():
    line.set_linewidth(3.5)

fig.tight_layout()
fig.savefig(FIG_DIR / "coffee_holdout_plot.png", dpi=300, bbox_inches="tight")
plt.show()

```



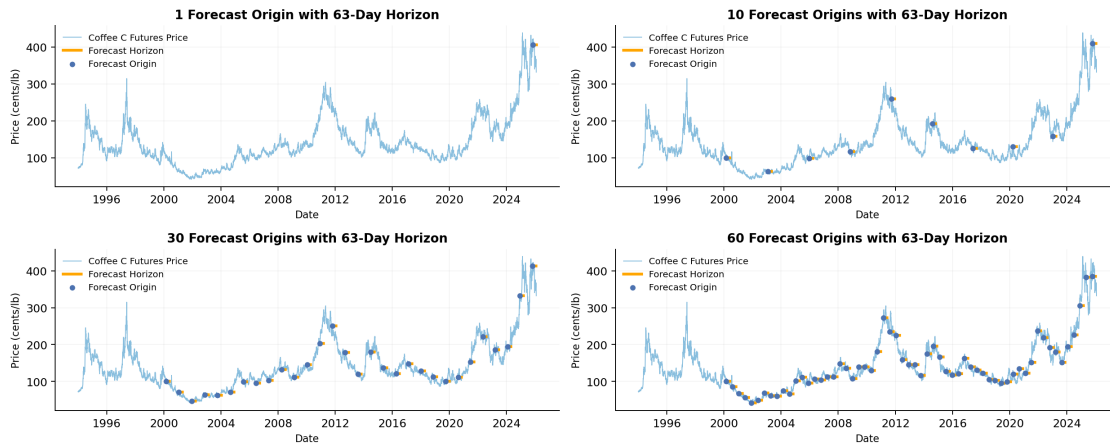
Reading this plot. Each model's 63-day forecast, overlaid with the realized path. Visually, everything looks sensible: the simple baselines cluster near the last observation, and the more flexible models (Prophet, Granite-TTM) make stronger directional claims. Granite-TTM in particular tracks the realized path more closely than anything else, which set our expectations going into the broader test.

1.6 5. Forecast origin coverage

With the single-window check behind us, we needed multiple origins to fairly cover the 32-year series. We started at 10 because it felt reasonable, but on second look the origins were too sparse to represent the full range of regimes in the data. We bumped it to 30 (better), then 60, close enough together to feel robust without crowding or overlapping. The plot below shows where each set of origins falls on the price history.

```
[7]: _ = plot_forecast_origin_coverage(
    df,
    scales=SCALES,
    save_path=FIG_DIR / "forecast_origin_coverage.png",
)
plt.show()
```

Rolling-Window Backtest Design: 1, 10, 30, 60 Forecast Origins



1.7 6. Run the rolling-window backtest

In practice we ran the rolling backtest scale by scale (10, then 30, then 60) as we iterated on the design. The cell below collapses that into one sequential run for reproducibility — it includes `scale=1`, which reproduces the §4 single-origin test.

`run_multi_scale_backtest` loops every (scale, model) combination and returns:

- `summary`: long-format DataFrame with one row per (origin, model, scale).
- `step_errors`: nested dict where `step_errors[scale][model_name]` is an ndarray of shape (num_origins, horizon) with absolute errors at each step. This per-step table is what feeds the Diebold-Mariano test in notebook 02.

```
[8]: summary, step_errors = run_multi_scale_backtest(
    df=df,
    models=models,
    scales=SCALES,
    context_len=CONTEXT_LEN,
    horizon=HORIZON,
    verbose=True,
)
print(f"\nTotal rows in summary: {len(summary)}")
summary.head()
```

Running 1 forecast origin(s)...

Naive
RWD
Theta
ETS
ARIMA
GARCH

Prophet
Ridge
RF
Granite-TTM

Running 10 forecast origin(s)...

Naive
RWD
Theta
ETS
ARIMA
GARCH
Prophet
Ridge
RF
Granite-TTM

Running 30 forecast origin(s)...

Naive
RWD
Theta
ETS
ARIMA
GARCH
Prophet
Ridge
RF
Granite-TTM

Running 60 forecast origin(s)...

Naive
RWD
Theta
ETS
ARIMA
GARCH
Prophet
Ridge
RF
Granite-TTM

Total rows in summary: 1010

```
[8]:      MAE      RMSE      MAPE      sMAPE      MASE  origin_id  origin_date  \
0  26.038889  29.875577  0.071331  0.068703  7.211687         1  2025-11-03
1  29.230761  35.042854  0.080769  0.076698  8.095703         1  2025-11-03
2  27.325129  31.970789  0.075154  0.071951  7.567922         1  2025-11-03
```

3	26.038889	29.875575	0.071331	0.068703	7.211687	1	2025-11-03
4	26.044184	29.926982	0.071368	0.068716	7.213154	1	2025-11-03

	model	scale
0	Naive	1
1	RWD	1
2	Theta	1
3	ETS	1
4	ARIMA	1

1.8 7. Export the two CSVs

Both files are stored in `results/csv/`. Notebook 02 (backtest stats) reads them directly and does not need to re-run the backtest. The visualizations in §9–§11 below also use this same data, but in-memory from the run we just completed.

```
[9]: summary_path = CSV_DIR / "summary_all_scales.csv"
summary.to_csv(summary_path, index=False)
print(f"Saved: {summary_path.relative_to(REPO_ROOT)}")

step_errors_df = step_errors_to_long_df(step_errors)
step_errors_path = CSV_DIR / "step_errors_all_scales.csv"
step_errors_df.to_csv(step_errors_path, index=False)
print(f"Saved: {step_errors_path.relative_to(REPO_ROOT)}")
↳ ({len(step_errors_df)} rows)
```

Saved: `results/csv/summary_all_scales.csv`

Saved: `results/csv/step_errors_all_scales.csv` (63630 rows)

1.9 8. The 60-origin leaderboard

The final ranking from the scale we settled on. The result was genuinely surprising: RWD on top, Granite TTM mid-pack despite the strong showing on the single window in §4. Whether these differences are statistically meaningful, or whether the top of the table is essentially a tie, is the question notebook 02 takes up with the Diebold-Mariano test and the Model Confidence Set.

```
[10]: leaderboard = (
    summary[summary["scale"] == 60]
    .groupby("model")[["MAE", "RMSE", "MASE", "sMAPE"]]
    .mean()
    .sort_values("MAE")
)
print("60-origin mean metrics (sorted by MAE):")
leaderboard.round(3)
```

60-origin mean metrics (sorted by MAE):

```
[10]:          MAE    RMSE    MASE  sMAPE
model
```

RWD	12.789	14.841	6.103	0.083
ETS	12.819	14.893	6.110	0.085
Theta	12.862	14.934	6.133	0.085
Naive	12.909	14.986	6.147	0.085
RF	13.132	15.188	6.282	0.087
ARIMA	13.239	15.385	6.283	0.085
GARCH	13.505	15.740	6.504	0.090
Granite-TTM	13.842	16.089	6.551	0.092
Ridge	14.682	17.160	6.872	0.101
Prophet	22.687	24.994	10.603	0.152

1.10 9. MAE across forecast-origin scales

Mean MAE per model at each of the four scales we ran (1, 10, 30, 60). As we add origins, the values stabilize — supporting our choice of 60 as a robust scale for the leaderboard. The x-axis is plotted as four equally-spaced ordinal steps so the early ramp from 1 to 10 isn't visually compressed.

```
[11]: mae_by_scale = summary.groupby(["scale", "model"])["MAE"].mean().reset_index()

# Map true scales to equally spaced positions so 1 -> 10 -> 30 -> 60
# reads as four ordinal steps rather than a squashed linear axis.
scale_positions = {s: i for i, s in enumerate(SCALES)}

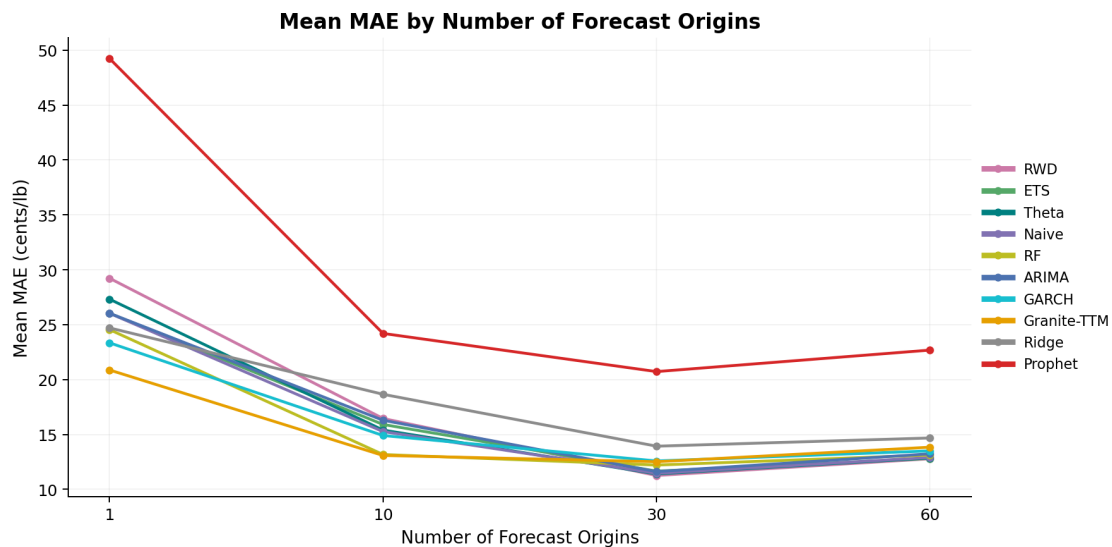
fig, ax = plt.subplots(figsize=(10, 5), dpi=180)
for name in PLOT_ORDER:
    model_data = mae_by_scale[mae_by_scale["model"] == name].
    ↪sort_values("scale")
    x_pos = [scale_positions[s] for s in model_data["scale"]]
    ax.plot(
        x_pos,
        model_data["MAE"],
        marker="o",
        color=COLOR_MAP.get(name, "#666666"),
        linewidth=1.8,
        markersize=4,
        label=name,
    )

ax.set_title("Mean MAE by Number of Forecast Origins", fontsize=13,
    ↪weight="semibold")
ax.set_xlabel("Number of Forecast Origins", fontsize=11)
ax.set_ylabel("Mean MAE (cents/lb)", fontsize=11)
ax.set_xticks(range(len(SCALES)))
ax.set_xticklabels(SCALES)
ax.grid(True, alpha=0.15)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
```

```

leg = ax.legend(loc="center left", bbox_to_anchor=(1, 0.5), frameon=False,
               ↪fontsize=9)
for line in leg.get_lines():
    line.set_linewidth(3.5)
fig.tight_layout()
fig.savefig(FIG_DIR / "mae_stabilization.png", dpi=300, bbox_inches="tight")
plt.show()

```



Reading this plot. The nine non-Prophet models converge to a tight cluster by scale = 30 and stay there at scale = 60. Prophet sits well above the cluster at every scale.

1.11 10. MAE distribution at 60 origins

The mean MAE sorts the models, but it can hide fat tails: a model that's average most of the time but occasionally explodes will look worse than it deserves, and vice versa. Boxplots show the full per-origin distribution: median, IQR, whiskers, and individual origins that fell outside the whiskers.

The chart below uses three design choices to keep the question crisp:

- **Sorted by median**, not mean, so visual order matches what the central line of each box shows.
- **Mean overlaid as a black diamond (◆)**, so the reader can compare median to mean inside each box. When they diverge, the error distribution is skewed.
- **Two-tone coloring** — the nine cluster models in neutral gray, Prophet (the model eliminated by MCS in nb02) in red — to make the cluster-vs-outlier picture obvious before any labels are read.

```
[12]: results_60 = summary[summary["scale"] == 60]
```

```

# Sort by MEDIAN (so visual order matches the box's central line).
# Worst on top: descending median + vert=False puts top at top of axis.
order = (
    results_60.groupby("model")["MAE"]
    .median()
    .sort_values(ascending=False)
    .index.tolist()
)
data = [results_60[results_60["model"] == m]["MAE"].values for m in order]
means = [float(np.mean(d)) for d in data]

# Two-tone coloring: cluster (top 9) in neutral gray; Prophet (the only
# MCS-eliminated model) in red. Emphasizes the cluster-vs-outlier story.
PROPHET_COLOR = "#E74C3C"
CLUSTER_COLOR = "#888888"
colors = [PROPHET_COLOR if m == "Prophet" else CLUSTER_COLOR for m in order]

fig, ax = plt.subplots(figsize=(10, 6), dpi=180)
bp = ax.boxplot(
    data,
    vert=False,
    patch_artist=True,
    medianprops=dict(color="black", linewidth=2),
    whiskerprops=dict(linewidth=1.2),
    capprops=dict(linewidth=1.2),
    flierprops=dict(marker="o", markersize=3, alpha=0.5),
    widths=0.55,
)
for patch, c in zip(bp["boxes"], colors):
    patch.set_facecolor(c)
    patch.set_alpha(0.7)
for flier, c in zip(bp["fliers"], colors):
    flier.set_markerfacecolor(c)
    flier.set_markeredgcolor(c)

# Mean markers (black diamonds) overlaid inside each box.
ax.scatter(
    means,
    range(1, len(order) + 1),
    marker="D",
    s=42,
    color="black",
    zorder=3,
    label="Mean ( )",
)

ax.set_xlim(0, 45)

```

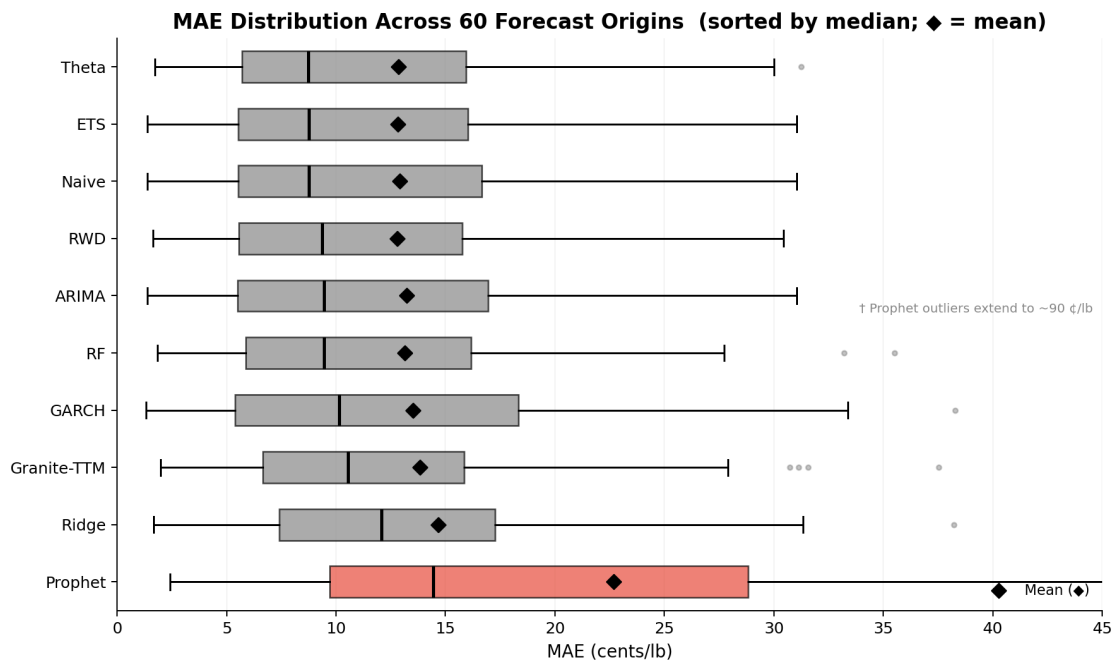
```

ax.annotate(
    "† Prophet outliers extend to ~90 ¢/lb",
    xy=(45, 0.5),
    xycoords=("data", "axes fraction"),
    ha="right",
    va="bottom",
    fontsize=8,
    color="#888",
    xytext=(-6, 6),
    textcoords="offset points",
)

ax.set_yticklabels(order, fontsize=10)
ax.set_title(
    "MAE Distribution Across 60 Forecast Origins (sorted by median; ♦ = mean)",
    fontsize=13,
    weight="semibold",
)

ax.set_xlabel("MAE (cents/lb)", fontsize=11)
ax.grid(True, axis="x", alpha=0.15)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.legend(loc="lower right", frameon=False, fontsize=9)
fig.tight_layout()
fig.savefig(FIG_DIR / "model_comparison_60.png", dpi=300, bbox_inches="tight")
plt.show()

```



Reading this plot. Two findings, one in each direction:

1. **The cluster overlaps heavily.** The IQRs of the nine non-Prophet models all sit on top of each other — the visual version of what MCS proved rigorously in notebook 02 (9 of 10 models survive in the MCS at $\alpha = 0.10$; only Prophet eliminated). The gray-vs-red coloring makes the eliminated-vs-retained dichotomy obvious before any labels are read.
2. **Mean and median don't always agree on rank.** For most cluster models the diamond (◆) sits near the median line, meaning the per-origin error distribution is roughly symmetric. For IBM Granite TTM the diamond sits noticeably right of the median, indicating right-skew — occasional bad windows pull the mean up. Prophet's mean is far right of its median for the same reason, much more pronounced; that skew (combined with the larger overall location) is one visual reason MCS eliminates it.

Within the cluster, IBM Granite TTM is the standout: its IQR is among the tightest, but the box sits shifted right of the leaders. So it's not noisy — just consistently a bit worse. The per-origin deep dive in §11 makes that pattern concrete.

1.12 11. Per-origin deep dive: IBM Granite TTM vs. RWD

Recall Granite TTM topped the single-window leaderboard back in §4. Across all 60 origins, the picture flips: it underperforms RWD almost everywhere. The interesting question is *how* it loses. Does it win often but occasionally collapse, or does it just underperform across the board? Plotting per-origin MAE for both models tells that story directly.

```
[13]: from matplotlib.lines import Line2D
      from scipy.ndimage import uniform_filter1d

      # Per-origin MAE = mean of per-step absolute errors over the 63-day horizon.
      s60 = step_errors_df[step_errors_df["scale"] == 60]
      rwd_mae = (
          s60[s60["model"] == "RWD"].groupby("origin_id")["error"].mean().
          ↪sort_index().values
      )
      gra_mae = (
          s60[s60["model"] == "Granite-TTM"]
          .groupby("origin_id")["error"]
          .mean()
          .sort_index()
          .values
      )
      origin_ids = np.arange(1, len(rwd_mae) + 1)

      # Smoothed trend to cut through origin-to-origin noise.
      W = max(5, len(rwd_mae) // 10)
      smooth_r = uniform_filter1d(rwd_mae.astype(float), size=W, mode="nearest")
      smooth_g = uniform_filter1d(gra_mae.astype(float), size=W, mode="nearest")
```

```

fig, ax = plt.subplots(figsize=(12, 5), dpi=180)

# Blue base layer: 0 to RWD trend (shared forecast difficulty).
ax.fill_between(origin_ids, 0, smooth_r, color="#AED6F1", alpha=0.7)
# Orange layer: only where Granite exceeds RWD (the extra error).
ax.fill_between(
    origin_ids,
    smooth_r,
    smooth_g,
    where=(smooth_g >= smooth_r),
    color="#FAD7A0",
    alpha=0.85,
)

ax.plot(origin_ids, smooth_r, color="#1A5276", linewidth=2.0, alpha=0.9)
ax.plot(origin_ids, smooth_g, color="#E67E22", linewidth=2.0, alpha=0.9)
ax.scatter(origin_ids, rwd_mae, color="#1A5276", s=40, zorder=5)
ax.scatter(origin_ids, gra_mae, color="#E67E22", s=40, zorder=5)

ax.set_title(
    "Per-Origin MAE: IBM Granite TTM vs RWD (60-Origin Scale)",
    fontsize=13,
    weight="semibold",
)

ax.set_xlabel("Forecast Origin", fontsize=11)
ax.set_ylabel("MAE (cents/lb)", fontsize=11)
ax.set_ylim(bottom=0)
ax.grid(True, ls="--", linewidth=0.5, alpha=0.5)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

legend_handles = [
    Line2D(
        [0],
        [0],
        marker="o",
        color="#E67E22",
        markerfacecolor="#E67E22",
        markersize=7,
        linewidth=4,
        alpha=0.7,
        label="IBM Granite TTM ( actual, smoothed)",
    ),
    Line2D(
        [0],
        [0],
        marker="o",

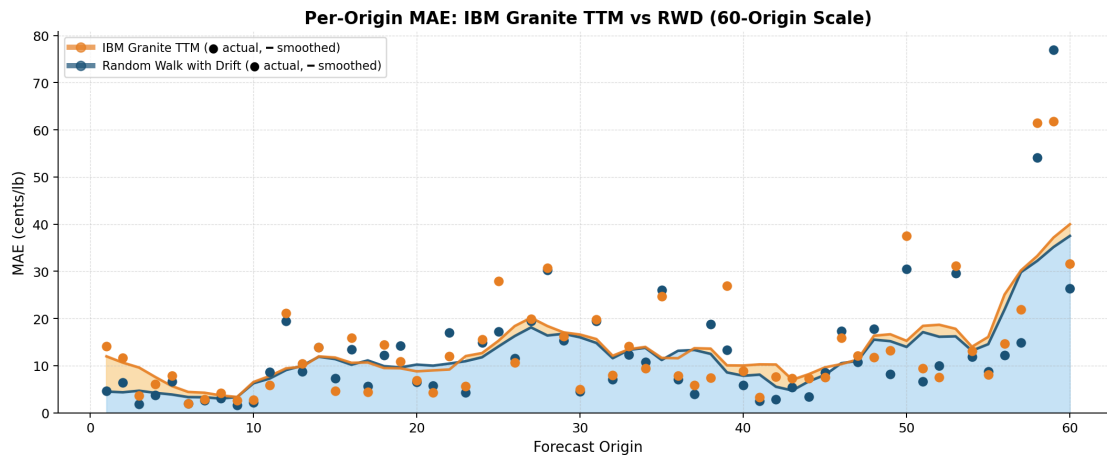
```

```

        color="#1A5276",
        markerfacecolor="#1A5276",
        markersize=7,
        linewidth=4,
        alpha=0.7,
        label="Random Walk with Drift ( actual, smoothed)",
    ),
]
ax.legend(handles=legend_handles, loc="upper left", frameon=True, fontsize=9)
fig.tight_layout()
fig.savefig(FIG_DIR / "IBM_vs_RWD.png", dpi=300, bbox_inches="tight")
plt.show()

granite_wins = int(np.sum(gra_mae < rwd_mae))
print(f"\nGranite beats RWD in {granite_wins} / {len(rwd_mae)} origins")
print(f"Mean MAE: Granite = {gra_mae.mean():.3f}, RWD = {rwd_mae.mean():.3f}")

```



Granite beats RWD in 16 / 60 origins
Mean MAE: Granite = 13.842, RWD = 12.789

Reading this plot. Granite has lower MAE than RWD at only 16 of 60 origins. The underperformance is not driven by a few outlier windows; the orange-shaded area (Granite's excess error above RWD) is nearly continuous, and the smoothed curves rarely cross. Neither the 32-year pretraining corpus nor the 1,536-day context window at each origin translate into lower point-forecast error on this series.

Next notebook: 02_backtest_stats.ipynb — Diebold-Mariano tests and the Model Confidence Set on the 60-origin results, to verify that the tight MAE cluster from §10 is an actual statistical tie rather than a ranking the sample size failed to resolve.